

A06 ENHANCED VENTILATION DESIGN | O (MAX : 3 PT)

Intent : Expel internally generated pollutants and improve air quality in the breathing zone through an increased supply of outdoor air or increased ventilation efficiency.

Summary : This WELL feature requires implementation of advanced ventilation strategies that can achieve higher air quality levels and thus benefit human health and productivity.

Issue : The majority of ventilation standards specify ventilation rates and other measures intended to provide indoor air quality that is merely "acceptable" to building users and reduces the risk of adverse health effects.¹ Even with proper ventilation designed to meet ventilation standards, the concentration of indoor pollutants can exceed concentrations found in outdoor air.^{2,3} Ventilation rates less than 10 L/s per person in all building types are associated with negative perception of air quality and actual health outcomes.⁴

Solutions : Scientific research suggests that an airflow rate significantly exceeding that recommended by standards is needed to minimize sick building syndrome symptoms and to improve human performance and productivity.^{5,6} Since it is difficult to test for every potential pollutant, and because carbon dioxide (CO₂) is easy to detect, CO₂ levels serve as a proxy for other indoor pollutants. A number of CO₂ studies suggest that the risk of sick building syndrome symptoms decreases significantly, when CO₂ concentrations are less than 800 ppm.⁴ One method for decreasing the CO₂ concentration experienced by occupants while minimizing additional energy use is demand-controlled ventilation, in which the delivery rate for outside air is directly linked to the measured CO₂ levels within the space.⁷ Displacement ventilation improves indoor air quality, by delivering fresh outdoor air at the floor level, leaving warmer polluted indoor air, such as CO₂ to be extracted above the height of the ventilation zone.⁸

PART 1 INCREASE OUTDOOR AIR SUPPLY (MAX : 2 PT)

For All Spaces:

Option 1: Increased air supply

For mechanically ventilated buildings, the following requirement is met in all occupiable spaces:

- Exceed outdoor air supply rates described in one of the ventilation guidelines listed in Feature A03 Part 1 by the percentages shown in the table below:

Tier	Thresholds	Points
1	30%	1
2	60%	2

OR

Option 2: Demand control ventilation

For mechanically ventilated buildings, the following requirements are met in at least 90% of regularly occupied spaces:

- A demand-controlled ventilation (DCV) system regulates the outdoor air ventilation rate to keep CO₂ levels less than the thresholds specified in the table below, at the maximum intended occupancy:

Tier	Threshold	OR	Threshold	Points
1	900 ppm	OR	500 ppm above outdoor levels	1
2	750 ppm	OR	350 ppm above outdoor levels	2

- Carbon dioxide is measured at the return air diffusers or in the breathing zone at least 1 m away from doors, windows, air supply diffusers or occupants. At least one sensor is used for each occupancy zone (or per air handling unit, if a single zone is served by multiple air handling units). If the occupancy density/pattern/usage is substantially different in two adjacent areas, each area must be considered a separate zone.

OR

Option 3: Enhanced natural ventilation

For naturally ventilated buildings, the following requirement is met:

- Implement an engineered natural ventilation system that is sufficient to keep CO₂ levels in the breathing zone of all regularly occupied spaces below the specified thresholds at the maximum intended occupancies:

Tier	Threshold	OR	Threshold	Points
1	900 ppm	OR	500 ppm above outdoor levels	1
2	750 ppm	OR	350 ppm above outdoor levels	2

OR

Option 4: Ventilation monitoring

One of the following carbon dioxide thresholds is met in occupiable spaces:

- 750 ppm or less.
- Not more than 350 ppm above outdoor levels.

Note : This Option is worth 2 points. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

Note : Mixed-mode projects must select the option corresponding to the ventilation mode (mechanical and natural ventilation) in operation for the majority of operating hours within the year.

PART 2 IMPROVE VENTILATION EFFECTIVENESS (MAX : 1 PT)

For All Spaces:

Option 1: Displacement ventilation system

The project uses a displacement ventilation system in at least 90% of regularly occupied spaces, with one of the following as a basis for design:

- a. ASHRAE Guidelines RP-949.⁹
- b. ASHRAE 62.1-2019, "Stratified Air Distribution Systems (Section 6.2.1.2.1).¹⁰
- c. REHVA Guidebook No. 01 (Displacement Ventilation in non-industrial premises).¹¹

OR

Option 2: Personalized ventilation system

At least 50% of workstations incorporate personalized ventilation systems that meet the following requirements:

- a. One of the following strategies is implemented to efficiently deliver air to occupants:
 1. Outdoor air is supplied in the breathing zone through vents incorporated into or mounted on furniture, and the return air diffusers within the space are located at least 2.8 m above the floor.¹⁰
 2. The system's zone air distribution effectiveness (E_z) is at least 1.20.¹⁰
- b. One of the following strategies is implemented to support thermal comfort:
 1. Outdoor air is supplied with an air speed of no greater than 0.25 m/s at the occupant's head.¹⁰
 2. The air speed from each outdoor air supply vent is individually controllable by occupants through a controller within the space, a mobile application or website.

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